



ArcGIS Blog

Analytics

ArcGIS Pro

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Use H3 to create multiresolution hexagon grids in ArcGIS Pro 3.1

By [Margaret Crawford](#)

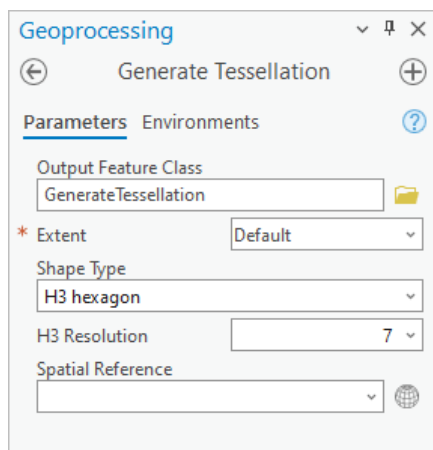
Approximate Reading Time: 8 minutes

In [ArcGIS Pro 3.1](#), the [Generate Tessellation](#) tool can now create hexagon grids using the open-source hierarchical spatial indexing system, H3. H3 hexagons were first introduced in ArcGIS Pro 3.0 by the [Business Analyst](#) team as an option for the licensed [Generate Grids and Hexagons](#) tool. Now, H3 hexagons are also available in the core [Generate Tessellation](#) geoprocessing tool!

Generate Tessellation tool

The [Generate Tessellation](#) tool creates a grid of repeating shapes over a given area. In previous versions of ArcGIS Pro, the tessellated shapes included triangles, squares, diamonds, hexagons (flat-top), and transverse hexagons (pointy-top). In [ArcGIS Pro 3.1](#), the [Generate Tessellation](#) tool continues to offer all the same tessellation shapes, with the addition of [H3 Hexagons](#).

H3 hexagons are available in the **Shape Type** parameter's drop-down list. You can set the average area of each H3 hexagon in the **H3 Resolution** parameter, where you can choose between 16 precomputed [H3 resolutions](#) (0 through 15). The bigger the resolution, the smaller the average area is for each hexagon—this results in more hexagons covering the same area.

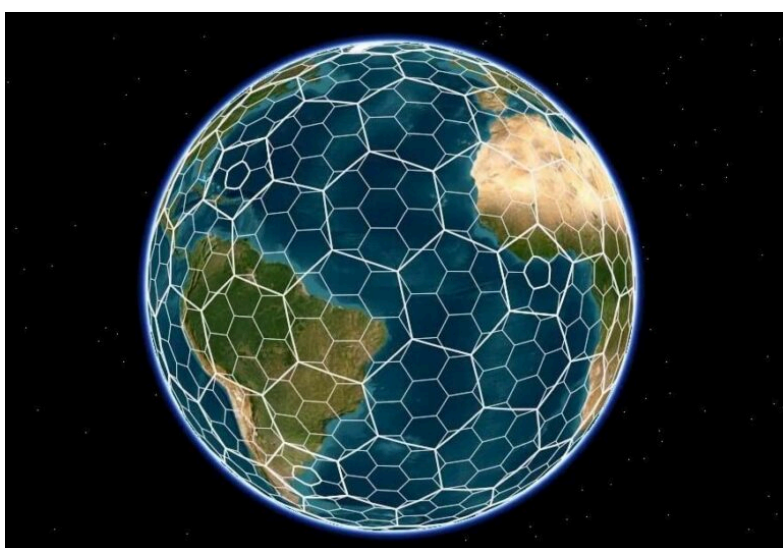


The Generate Tessellation tool

If you are unfamiliar with H3 hexagons, this blog will briefly introduce them. If you are already familiar with H3 Hexagons, keep reading to see how to use H3 hexagons for [spatial analysis](#).

What is H3?

H3 is a hierarchical indexing system [developed by Uber](#) that uses hexagons to tile the Earth's surface. H3 builds its grid of hexagons over an icosahedron: a shape with 20 triangular faces and 12 vertices. Since it is impossible to tile an icosahedron with only hexagons, each resolution needs 12 pentagons to complete the grid. A hierarchical indexing grid means each hexagon can subdivide into sub-unit hexagons. Therefore, H3 can index any location on Earth, starting from an average area of 4,357,449.41 km² (resolution 0) down to 0.895 m² (resolution 15).



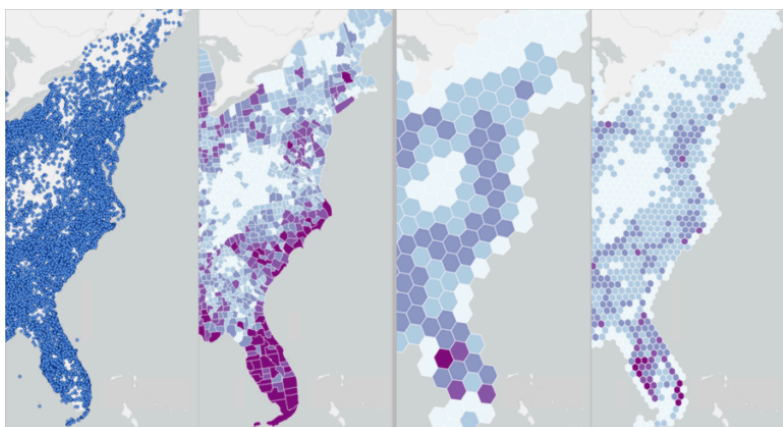
H3 is a global grid system that partitions the Earth into hexagons and 12 pentagons

H3 has been written about in detail in other places, [including the H3 documentation](#). But you might be asking, why should I care about a new type of hexagon grid?

Why use H3 Hexagons for Spatial Analysis?

Spatial binning or aggregation is an excellent way to analyze patterns in large point datasets. There are several advantages to using hexagonal grids for aggregation and summarization rather than administrative boundaries like states, counties, or block groups.

Using a uniform grid of hexagons, you can display raw count statistics. Administrative boundaries are irregular in size and shape, which may cause size bias where larger objects appear more prominent and require normalization for numeric attributes. Also, unlike administrative boundaries, hexagons do not change over time, meaning you can use them to compare data over different time horizons. In particular, H3 hexagons are great because they are built over a model of the Earth, meaning their position remains consistent at each resolution. Therefore, you can use H3 hexagons as a standardized grid across multiple scales of analyses and projects.



From left to right: a point dataset, and points binned using counties, H3 hexagons at resolution 3, and H3 hexagons at resolution 4

[Lean more about the advantages of hexagon grids in spatial analysis.](#)

Example: Analyzing tornado data using H3 hexagons

This example uses the initial point locations of tornadoes from 1950 to 2021 available on the [National Weather Service SVRGIS page](#). The data includes each tornado's

start and end location, the year it occurred, and the estimated property loss (in millions of dollars).

OBJECTID	year	property loss	start lat.	start long.	end lat.	end long.
1	1950	4	36.73	-102.52	36.88	-102.3
2	1950	5	34.17	-78.6	34.1701	-78.59989
3	1950	5	37.37	-87.2	37.3701	-87.1999

Our goal is to visualize the impact of tornadoes from the last decade (2011 to 2021) over the lower 48 United States.

Create a grid of H3 Hexagons using the Generate Tessellation tool

Using the [Generate Tessellation](#) tool, set the following parameters:

(1) Specify the **Output Feature Class** as

`memory\h3_hexagons`.

Note: Because this step is intermediate, [writing data to memory](#) provides the fastest performance by creating a temporary hexagon layer.

(2) Position your map over the lower 48 states, then set **Extent** to Current Display Extent.

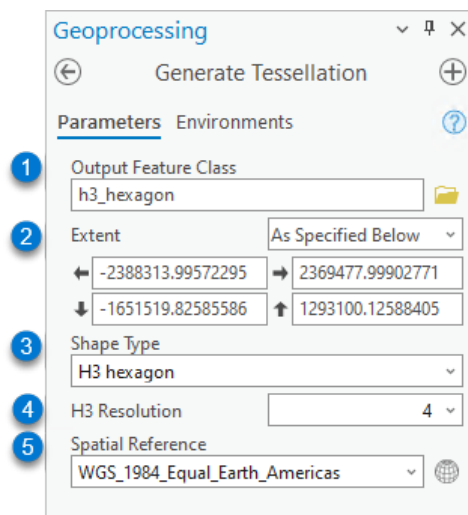
Note: This will populate the extent coordinate boxes with your map's current min and max X/ Y coordinates.

(3) Specify the **Shape Type** as *H3 hexagon*

(4) Set the **H3 Resolution** to 4

(5) Specify the desired **Spatial Reference**.

Note: We used the [Equal Earth](#) Americas projection to minimize area distortion. Using a UTM or State Plane projected coordinate system would be inappropriate for a large continental area grid. Also, you should avoid generating a large grid using [Web Mercator](#).



Generate Tessellation tool

Run the **Generate Tessellation** tool.

The resulting feature class contains a grid of H3 hexagons with an **H3 Resolution** of 4. The average area of each hexagon is roughly 1,770.3 square kilometers, where some of the hexagons are smaller or larger than this average size.



Generate Tessellation tool output with H3 hexagons at resolution 4

Select only the hexagons that intersect the United States

Notice that the tool's output is a large grid covering the whole map extent.

For this example, we want to limit our area of interest to the lower 48 United States:

Using **Select by Location**  set the following parameters:

(1) Set **Input Features** as *h3_hexagon*

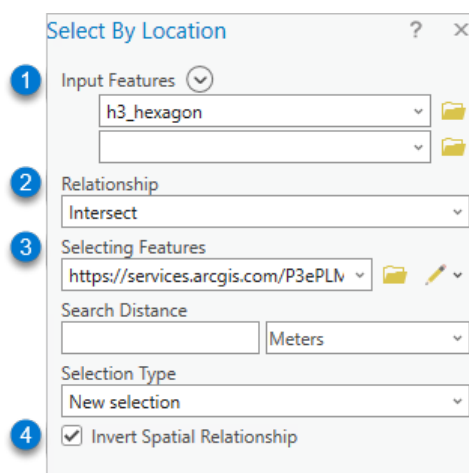
(2) Set **Relationship** to *intersect*

(3) Copy and paste the following URL for the **Selecting Features**:

https://services.arcgis.com/P3ePLMYs2RVChkJx/arcgis/rest/services/USA_Boundaries_2021/FeatureServer/

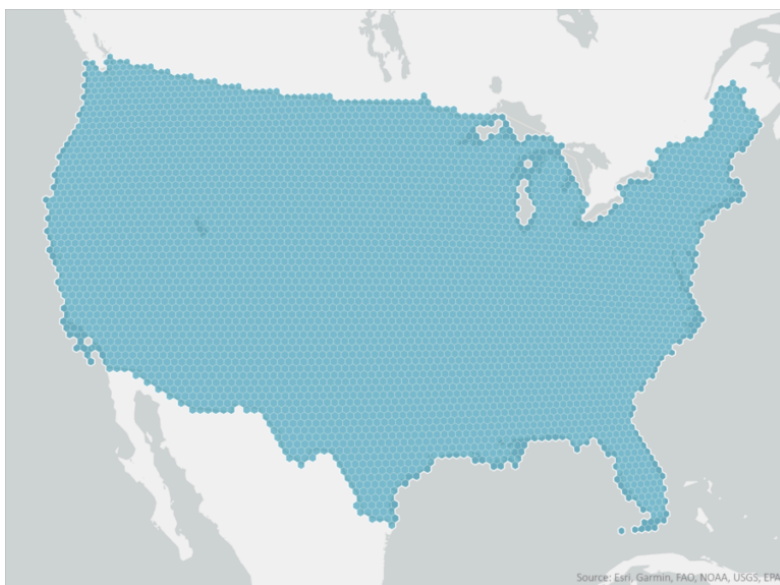
Note: This is a feature layer of USA Boundaries from the ArcGIS Living Atlas; you can use the feature service path as inputs to geoprocessing tools.

(4) Check **Invert Spatial Relationship** to select all hexagons that do NOT intersect the USA.



Select by Location

(5) Use the **Delete Features** tool to remove all selected features from the H3_hexagon layer



A grid of H3 Hexagons at resolution 4 over the lower 48 states.

Summarize the tornadoes within each H3 hexagon

Lastly, we will use the Summarize Within tool:

(1) For **Input Polygons**, choose the H3 hexagon layer of the United States

(2) Set **Input Summary Features** as the tornado layer

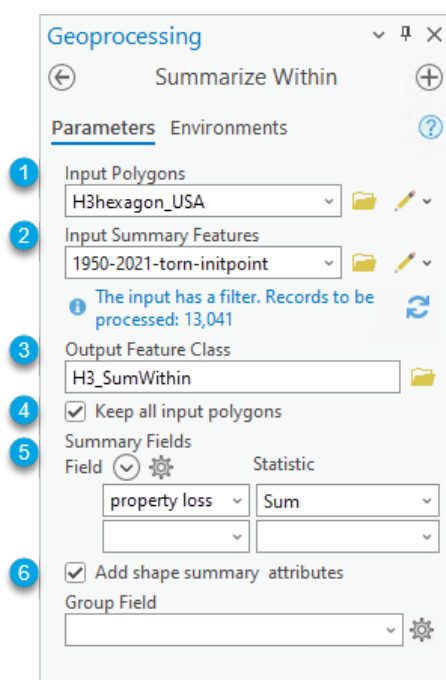
Note: the tip indicates a filter is applied to the layer. We applied a [Definition Query](#) to use only tornadoes where the `year >= 2011`

(3) Specify the **Output Feature Class**

(4) Under **Summary Fields**, use *property loss* for the **Field** and *sum* for the **Statistic**

(5) Leave **Keep all input polygons** checked

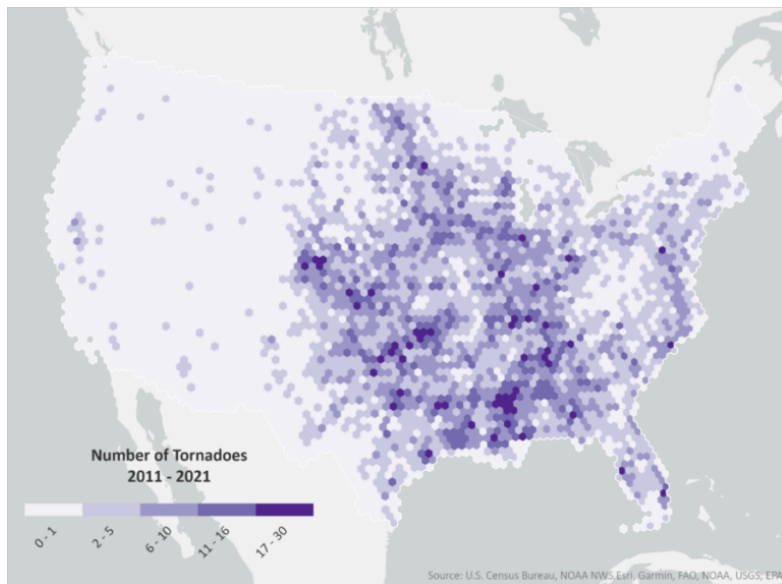
(6) Leave **Add shape summary attributes** checked to include a count of the number of points in each hexagon.



Summarize Within tool

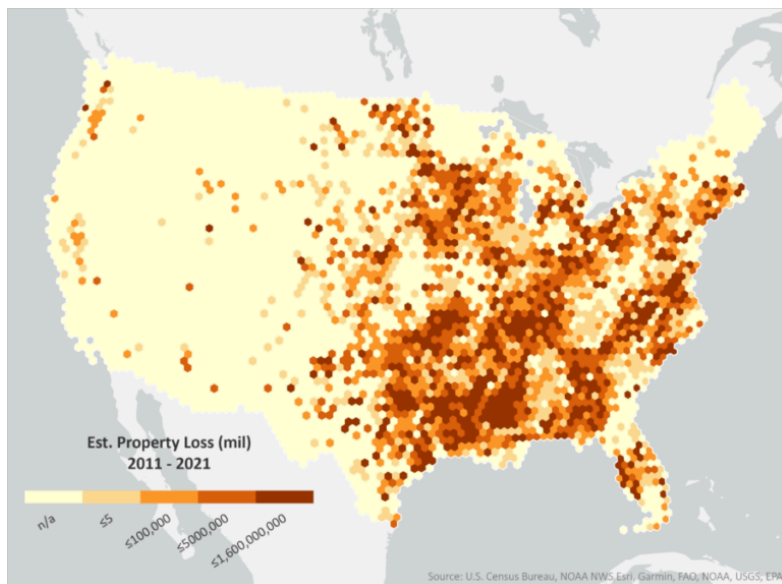
After running the **Summarize Within** tool, the resulting feature class will include the following:

(1) The raw counts of tornadoes per H3 hexagon



Number of tornadoes from 2011 to 2021

(2) The total estimated property damage per H3 hexagon.



Estimated property loss (in millions of dollars) caused by tornadoes from 2011 to 2021

Summary

H3 is a hexagonal hierarchical spatial indexing system developed by Uber. In [ArcGIS Pro 3.1](#), H3 hexagons are available in the [Generate Tessellation](#) geoprocessing tool. This tool creates a uniform grid of repeating shapes over any given area. You can use H3 hexagons as a standard grid for visualizing and comparing spatial patterns across multiple scales of analysis and locations. We hope you try using H3 hexagons in your next project!

Sources

- Sahr, k., (2011). HEXAGONAL DISCRETE GLOBAL GRID SYSTEMS FOR GEOSPATIAL COMPUTING. *Archives of Photogrammetry, Cartography and Remote Sensing, Vol. 22, pp. 363-376.*
- Uber Engineering. (2018, November 19). *[Uber Open Summit 2018] Hierarchical Hexagons in Depth*. [Video]. Youtube. <https://www.youtube.com/watch?v=U1LoSqvlM2w>
- Uber Technologies, Inc. (n.d.). H3. Retrieved February 10, 2023, from <https://h3geo.org/>

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Margaret is a Product Engineer on the ArcGIS Pro Geoprocessing team. Her passion for GIS began with her interest in human-environmental relationships, where her graduate thesis studied the growing impact of cascading disasters caused by sea level rise. Margaret leverages her foundation in research methodologies and spatial analysis techniques in her role at Esri, where she creates written content and demonstrations on new features in ArcGIS Pro. She also utilizes user feedback to refine and redesign the geoprocessing interface. In her personal life, Margaret is a loving cat parent to three beautiful cats.

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